



SUBHAN IRSYADUDDIEN ALHAQ

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Pasar Minggu, South Jakarta

Mathematics student at Universitas Indonesia with a strong interest in data science and analytics. Experienced in working with data using Python and SQL, from data cleaning and structuring to modeling. Strong in analytical thinking, problem decomposition, and delivering reproducible results.

EDUCATION

Universitas Indonesia

Bachelor of Mathematics, GPA: 3.51/4.00

Depok, West Java

Aug 2023 - Aug 2027 (Expected)

- Relevant Coursework: Mathematical Modeling, Data Science, Linear Models, Data Structures.

NON FORMAL EDUCATION

Mastering AI Engineer

Non Formal Education

Subjects Studied: Machine Learning, Deep Learning, Model Usage, Database, Computer Vision, NLP, LangChain, MLOps.

WORK EXPERIENCE

Mathematics Tutor - Tutorin Math

May 2025 - Present

Tutoring & Academic Preparation

- Designed and delivered 20+ learning modules for dozens of students across multiple programs: The Calculus Class, 100 Problems to Calculus Mastery, TKA, and CPNS TIU.
- Provided structured explanations, guided practice, and targeted feedback to strengthen students' calculus fundamentals and problem solving speed.
- Adapted teaching strategies to students' skill levels by diagnosing common errors and creating focused drills to improve accuracy and consistency.

Teaching Assistant - Universitas Indonesia

Sep 2025 - Dec 2025

Department of Mathematics

- Supported research-based learning activities by assisting course delivery for 40+ students.
- Graded and reviewed 6 student assignments to ensure consistent evaluation and timely feedback.
- Coordinated with the TA team to align rubrics and maintain grading consistency.

PROJECTS

Cloud Data Security Risk Analysis — Mathematical Modeling Course

Aug 2025 – Dec 2025

Data Science / Cybersecurity Analytics

Analyzed cloud storage security risk by modeling user behavior from activity logs and detecting anomalous usage patterns using unsupervised learning.

- Transformed raw cloud activity logs into daily per-user behavioral features by grouping and engineering event count/ratio features.
- Built an unsupervised anomaly detection pipeline using Isolation Forest to generate anomaly scores as risk indicators for suspicious user-day activity.
- Interpreted model decisions using Tree SHAP to identify the most influential behavior signals, and evaluated results using confusion-matrix based metrics.

Nutritional Status Classification (Balita) — Data Science Course

[Aug 2024 – Dec 2024]

Data Scientist

Multi-class classification project to predict toddler nutritional status (stunting risk) using anthropometric features, comparing Random Forest and ANN (MLPClassifier) with imbalance handling and hyperparameter tuning.

- Built a supervised learning pipeline on a 4-class label scheme (severely stunting, stunting, normal, tall) using key features: age (months), gender, and height (cm).
- Improved data quality and class balance via outlier handling and SMOTE; tuned models with GridSearchCV and evaluated with Accuracy, ROC-AUC, and Confusion Matrix.
- Achieved **~99% classification accuracy** after preprocessing, SMOTE, and tuning (reported in the project report/README).

House Price Regression Analysis — Linear Models Course

Aug 2024 – Dec 2024

Data Analyst (Regression Modeling)

Built and validated a multiple linear regression model to predict house prices using 1,000 observations and 7 predictors.

- Performed exploratory analysis (scatter plots, correlation heatmap) and data validation to confirm no missing values before modeling.
- Proposed two regression models and applied stepwise regression + nested-model ANOVA to select a simpler best model by removing non-significant predictors.
- Verified regression assumptions (linearity, independence via Durbin–Watson, homoskedasticity via Breusch–Pagan, normality via Shapiro–Wilk, and multicollinearity via VIF) and validated performance using cross-validation.

ORGANISATIONAL EXPERIENCE

UI Science Olympiad

Depok Jun 2024 - Nov 2024

Competition Coordinator

A national science competition organized by BEM FMIPA Universitas Indonesia to foster interest and talent in science among students.

- Managed three competition branches: Scientific Poster, Critical Essay, and Case Study.
- Developed technical guidelines, assessment criteria, and event timelines.
- Coordinated with judges, participants, and divisions to ensure smooth execution.
- Contributed to an event participated in by over 7,000 students nationwide.

Ksatria MIPA

Jul 2024 - Nov 2024

Treasurer Controller

A student organization that prepares FMIPA UI students for academic competitions.

- Managed and monitored financial transactions and budgeting.
- Ensured financial activities followed regulations and procedures.
- Compiled and documented financial reports for transparency.
- Coordinated with divisions to align fund allocations with competition needs.

Himpunan Mahasiswa Departemen (HMD) Matematika

Depok Apr 2025 - Present

Head of Academic Division

The Mathematics Student Association of FMIPA UI supports academic, social, and professional development for students.

- Planned and supervised academic programs such as seminars, training, and workshops.
- Coordinated with faculty for competitions organized by HMDM.
- Mentored division staff to ensure program effectiveness.
- Initiated inter-organizational collaborations to improve academic engagement.

SKILLS, ACHIEVEMENTS & OTHER EXPERIENCE

Programming & Tools: Python, SQL, Pandas, NumPy, Scikit-learn, Keras, Matplotlib, Seaborn, Streamlit, Google Colab, Google Data Studio, Git, VS Code, FastAPI, Selenium.

Data Science & Analytics: Data Cleaning, Data Visualization, Machine Learning, Statistical Analysis, Time Series Analysis, Natural Language Processing (NLP), Classification & Clustering, Sentiment Analysis.

Software & Productivity Tools: Google Workspace (Docs, Slides, Sheets, Form, Drive), Canva, Excel.

Soft Skills: Problem Solving, Critical Thinking, Leadership, Project Management, Communication.